

DHRUV PATEL

ENTERPRISE AI & DATA ENGINEERING, PRINCIPAL

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PROFILE

Hands-on AI/ML engineer who writes and ships LLM-agent and machine learning systems in Python, with 11+ years building production data and ML platforms in Fortune 100 environments. Currently a Sr. Forward Deployed Engineer at Sherwin-Williams, building the Freight Rate, Batch Size Rate and demand-forecasting models behind pricing. Outside work I design and operate a 15+ agent autonomous trading and research system with Claude Code and MCP integrations, where every agent output is graded against statistical baselines before it is trusted: evaluation harnesses, guardrails, permission boundaries, kill switches and monitoring, written and debugged by hand.

Earlier work spans Generative AI and LLM solutions (RAG and vector-database patterns) and developer-productivity tools, MLOps and model monitoring, explainability and responsible AI (Global Payments), TensorFlow and spaCy NLP models (Compliance.AI), and event-driven data platforms (Kafka/Confluent, Spark, Databricks, Snowflake, ClickHouse) on AWS, Azure and GCP. Comfortable as the engineer embedded with business and external stakeholders, from first prototype through production.

CORE COMPETENCIES

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| • Large Language Models (LLMs) | • Claude Code / AI Coding Tools | • AutoGluon / TensorFlow / Keras | • Snowflake |
| • Generative AI | • Machine Learning | • MLOps & CI/CD | • ClickHouse |
| • RAG (Retrieval-Augmented Generation) | • LLM Applications | • Vertex AI / BigQuery ML | • Delta Lake |
| • Vector Databases | • NLP (spaCy, TensorFlow) | • Python | • AWS / Azure / GCP |
| • LLM Agents & Tool Use | • Model Monitoring & Drift | • Demand Forecasting | • Terraform / IaC |
| • MCP Servers & Integrations | • Explainability & Responsible AI | • Pricing & Freight Rate Modeling | • Docker / Kubernetes |
| • Prompt & Output Evaluation | • Node.js | • Forward Deployed Engineering | • GitOps |
| • LLM Guardrails & Permissions | • SQL | • Apache Kafka / Confluent | • Informatica MDM / Reltio |
| | • Scala / Spark | • Databricks | • Azure RBAC / Key Vault |
| | • Apache Airflow | | • SOX / GDPR |
| | | | • Collibra / Purview |

HANDS-ON LLM & AGENT ENGINEERING

- **Autonomous trading & research ecosystem (15+ agents, live):** Built and operate in Python and Node.js: a shared SQLite 15m bar store (Alpaca IEX + Yahoo fallback); signal agents (SuperTrend flip watcher on 75 symbols, 11-sector IV spread scanner, 102-coin crypto monitor, Kalshi BTC caller); research tools (paper-lab for BTC/ETH and 4 CME micros with no broker path, sector-breadth study, real-estate digest); all on launchd with a watchdog that repairs missed runs against a NYSE calendar. *Source: github.com/darup67 (trade-core, market-lab, kalshi-btc-agent).*
- **Statistical evidence gating:** Every signal kind (equity flip, IV spread, crypto pick, coin launch) is graded net of costs with Wilson lower bounds (≥ 30 signals, ≥ 10 days); only PROVEN kinds fire live alerts, the rest are logged and withheld.
- **Tool use over MCP:** Wired agents to MCP servers for brokerage (Robinhood Agentic MCP: read-only scopes, confirm-before-order rule), charting (TradingView 84-tool MCP), email and job-search data. Enforced permission boundaries in code: credentials kept in OS keychain, never in prompts or logs.
- **LLM output in shadow mode:** Integrated a hosted LLM headline-judging service to label topic, direction and materiality, storing each judgment beside the outcome so it is graded before it can influence any decision.
- **Models with guardrails:** Trained AutoGluon classifiers with held-out testing, minimum-AUC gates and kill switches that turn a model off when live results decay; diagnosed and fixed failures such as leaky cross-validation folds and stale features.
- **Text signals:** Built a sentiment pipeline over StockTwits tags, news, Reddit, CoinGecko and Telegram with a crypto lexicon and negation handling, logged each reading with the price, and tested it against what followed instead of assuming it predicts.

PROFESSIONAL EXPERIENCE

Sr. Forward Deployed Engineer | SHERWIN-WILLIAMS

Jun 2026 — Present

Forward Deployed Engineering

- **Forward Deployed Delivery:** Work directly with internal business stakeholders and external partners to turn business problems into deployed tools, technology and AI/ML-driven platforms that the business consumes directly and that support revenue generation.
- **Stakeholder Deliverables:** Gather requirements, define deliverables and drive them to production with business users, engineering and external stakeholders, and support adoption after release.

Pricing Structure & Demand Forecasting ML Models

- **Freight Rate Model (FRM):** Build and operate the Freight Rate Model, a machine learning model that estimates freight rates and supplies the freight component of the pricing structure, covering feature engineering, training, validation, deployment and ongoing performance monitoring.
- **Batch Size Rate (BSR) Model:** Build and operate the Batch Size Rate model, a machine learning model that sets rates by batch size and feeds the pricing structure, owning the full model lifecycle from data preparation through production release and monitoring.
- **Demand Forecasting:** Design, train and productionize machine learning models that forecast demand from historical data, supporting planning and inventory decisions and giving pricing a forward view of volume.
- **Pricing Structure:** Combine freight-rate, batch-size-rate and demand-forecast outputs so the pricing structure is built from data-driven, repeatable models instead of manual rate setting.

Principal AI/ML Engineer | T-MOBILE

Jun 2022 — Dec 2025

AI/ML Platform Engineering & Delivery

- **ML Delivery:** Led 25+ engineers and data scientists (budget over \$15M) to architect and deploy machine learning solutions for telecommunications operations; AI/ML adoption across 15+ business units saved \$2.3M a year through end-to-end automation.
- **Forward Deployed Delivery:** Worked directly with internal business units and external vendors and partners to scope, build and deliver tools, technology and AI/ML-driven platforms for business consumption and revenue generation, translating stakeholder needs into production deliverables.
- **CI/CD & MLOps:** Optimized deployment pipelines and integrated automated testing, achieving a 40% reduction in model deployment cycles while sustaining 99.8% system uptime.
- **Cloud Infrastructure:** Delivered trusted analytics via semantic data products and event-driven architectures across AWS and Microsoft Azure, leveraging GitOps and Infrastructure-as-Code (IaC) for scalable environment management.
- **High-Scale Microservices:** Managed the end-to-end lifecycle for MDM/RDM platforms through 12 major feature releases, supporting a microservices architecture that handled 10 million daily transactions for 50,000+ internal users.
- **Golden Record & Matching:** Built Golden Record solutions with fuzzy logic and machine learning matching algorithms consolidating 30+ sources: data quality up 35%, data reconciliation time down 60%, manual corrections down 45%.
- **Real-time Data Fabric:** Built integration patterns with Apache Kafka and REST APIs to process 500GB+ of daily data, improving data discovery 50% through automated lineage tracking.
- **Security & Migration:** Designed security architecture for the migration of 200+ applications and databases to Azure (assets over \$100M); automated data classification and Privileged Identity Management cut data exposure incidents 80% and unauthorized access 95%. Put SOX and GDPR controls around enterprise ML and data platforms.

Senior ML Data Engineer | GLOBAL PAYMENTS

Dec 2019 — May 2022

Generative AI, MLOps & Responsible AI

- **Forward Deployed Delivery:** Partnered with internal teams and external stakeholders to define, build and deliver tools, technology and AI/ML-driven platforms for business consumption and revenue generation, from requirements through production adoption.
- **Generative AI / LLM Solutions:** Developed Generative AI and LLM-based solutions, including internal developer-productivity tools that streamline engineering workflows.
- **MLOps & Model Monitoring:** Implemented MLOps practices and model monitoring for production ML models, tracking model health and performance after release.

- **Explainability & Responsible AI:** Applied model explainability and responsible-AI practices so automated decisions could be reviewed, audited and trusted.

Data Platform, Cloud & Machine Learning

- **High-Scale Pipeline Design:** Architected ETL pipelines processing \$50B+ annual transaction volumes using Python, Scala, and Apache Spark, migrating core workflows to GCP to improve system resiliency.
- **GCP Infrastructure Management:** Spearheaded the transition from AWS (S3, EMR) to GCP-native services, utilizing BigQuery and Cloud Storage to achieve a 45% reduction in query response times.
- **AI/ML Development (GCP):** Engineered end-to-end machine learning lifecycles within Vertex AI, deploying models for predictive analytics and automated decision-making.
- **Advanced Fraud Detection:** Enhanced real-time fraud detection systems to 98.5% accuracy and a 25% reduction in false positives by implementing BigQuery ML and streaming data transformations.
- **Cost Optimization & Migration:** Led a strategic migration initiative from legacy on-prem/AWS systems to GCP, realizing a 30% reduction in annual infrastructure overhead through resource rightsizing and BigQuery slot optimization.
- **LookerBI & Business Intelligence:** Orchestrated the integration between back-end infrastructure and BI systems, leveraging LookerBI to build automated reporting layers and real-time executive dashboards.
- **Data Mappings & Preprocessing:** Developed complex data mappings and schema definitions to align disparate financial datasets and optimized high-velocity data streams, preparing massive datasets for downstream AI/ML consumption while ensuring integrity across the enterprise data lake.
- **Delivery & Reliability:** Directed a cross-functional engineering team using Agile/Scrum, accelerating delivery 20% through sprint planning and CI/CD automation; scaled cloud-native back-end systems for high availability in mission-critical financial processing.

Master ML Data Engineer (MDM/RDM) | NOVELIS

Mar 2019 — Sep 2019

- **MDM/RDM Hub:** Implemented Informatica MDM hub for 15 global manufacturing sites, improving data quality from 78% to 96%.
- **Automation & Governance:** Developed Python/SQL workflows processing 2M+ records monthly; established governance processes and trained over 25 business users.

Data Engineer / ML Engineer | COMPLIANCE.AI

Sep 2017 — Feb 2018

- **NLP ML Models:** Developed TensorFlow and spaCy models for regulatory compliance, reaching 92% accuracy on document classification.
- **System Performance:** Built Airflow/Docker pipelines and REST APIs with 99.9% uptime and <100ms response times.

EDUCATION

Georgia Institute of Technology — BS in Computer/Information Technology Administration & Management	May 2019
University of Georgia — BBA in Actuarial Science, Risk Management & Insurance	May 2014

CERTIFICATIONS

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| • MIT Professional Certificate in ML & AI | • AWS ML Specialty |
| • IBM AI Engineering Professional | • Azure AI Engineer Associate (AI-102) |
| • Google TensorFlow Developer | • Google Cloud Professional Data Engineer |
| • DeepLearning.AI ML & Deep Learning Specializations | |